

Advanced Materials and Additive Manufacturing for Phase Change Thermal Energy Storage and Management: A Review

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Phase change materials (PCMs) can enhance the performance of energy systems by time shifting or reducing peak thermal loads. The effectiveness of a PCM is defined by its energy and power density—the total available storage capacity (kWh m^{-3}) and how fast it can be accessed (kW m^{-3}). These are influenced by both material properties as well as geometry of the energy systems; however, prior efforts have primarily focused on improving material properties, namely, maximizing latent heat of fusion and increasing thermal conductivity. The latter is often at the expense of the former. Advanced manufacturing techniques hold tremendous potential to enable co-optimization of material properties and device geometry, while potentially reducing material waste and manufacturing time. There is an emerging body of research focused on additive manufacturing of PCM composites and devices for thermal energy storage (TES) and thermal management. In this article, the fundamentals and applications of PCMs are reviewed and recent additive manufacturing advances in latent heat TES for both the PCM composite and associated heat exchanger are discussed. A forward-looking perspective on the future and potential of PCM additive manufacturing for TES and thermal management is provided.

1. Introduction

There is an urgent need to decarbonize our energy systems to mitigate the effects of climate change.^[1] Doing so will require the rapid deployment of clean energy generation such as solar and wind.^[2] However, the inherent intermittency of such sources requires co-deployment of scalable, affordable, and sustainable energy storage technologies.^[3] Over the years, several methods to store various forms of energy have been developed, each with its own level of readiness, advantages, and drawbacks.^[4–7] Of these options, thermal energy storage (TES) systems are of particular interest because of their potential for low costs and long lifetimes. Materials used for TES can be classified into three categories: 1) sensible, 2) latent, and 3) thermochemical. Many authors have presented comparisons as well as advantages and disadvantages of sensible,^[8]

latent,^[9,10] and thermochemical energy storage.^[11–13]

Latent heat storage materials, also known as phase change materials (PCMs), have great potential for a variety of thermal management applications because of their ability to store heat over a near constant temperature.^[14] Heat can be stored in PCMs using a variety of phase transitions, including gas-liquid, solid-gas, solid-solid, and solid-liquid. PCMs involving a gas phase are generally not of great interest to the TES community, as they suffer from very low energy density because the gaseous phase occupies a large volume.^[15] Solid-solid phase transitions typically occur as a material undergoes a transition from one crystalline phase (polymorph) to another, or from a crystalline to amorphous phase, and consequently are limited in energy density.^[16] As such, solid-liquid PCMs are most compelling in terms of energy density and will be the primary focus of this review.

The effectiveness of a PCM is defined by its energy and power density—the total available storage capacity (kWh m^{-3}) and how fast it can be accessed (kW m^{-3}) per unit volume. These properties are influenced by both material properties and device geometry. Most prior works focused on improving material properties by maximizing the latent heat of fusion while increasing thermal conductivity.^[17–19] The latter is often at the expense of the former,

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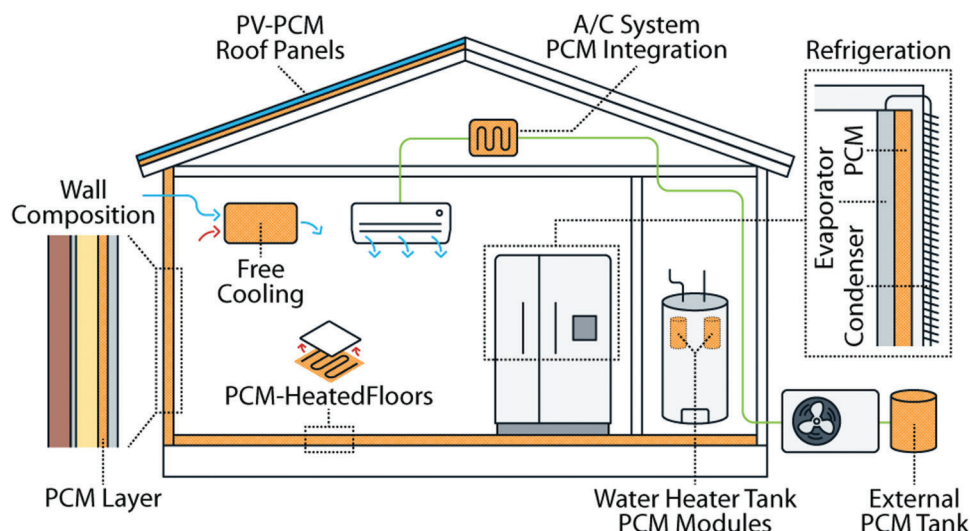


Figure 1. Potential implementations of TES within building applications.

as most efforts attempt to improve thermal conductivity by combining a thermally conductive material with the PCM to create a composite. The thermally conductive material does not participate in phase change, and thus displaces PCM volume. Woods et al. showed that maximizing energy and power density in a PCM device can be achieved by minimizing internal thermal resistances, which can be done by increasing thermal conductivity, increasing heat transfer surface area, and reducing heat transfer length scales.^[20] These can all be pursued effectively using advanced manufacturing techniques, creating tremendous potential to co-optimize the material properties and device geometry, while potentially reducing material waste and manufacturing time. There is an emerging body of research focused on additive manufacturing of PCM composites and devices for TES. In this review article, first, various applications and solid-liquid PCMs with transition temperature between -50 and 400 °C are reviewed. Second, a comprehensive review of recent additive manufacturing (AM) enabled advances in latent heat TES that have the potential to transform the field by enabling co-optimization of material properties and device geometry is provided. Lastly, a forward-looking perspective on potential future directions of advanced materials and additive manufacturing of PCM composites for TES and thermal management is provided.

1.1. Thermal Energy Storage Applications

Thermal energy plays a central role in many applications, from generating power to conditioning living spaces. TES devices can be used in any system that generates or transfers thermal energy, where the benefit of storage will depend on the application. Typical benefits of TES include energy efficiency improvements, emissions reduction, load shaving/shifting, and thermal management – the ability of the PCM to prevent operation outside a system's safe or comfortable operating temperature range.

One common application of TES is in building thermal systems. In fact, sensible thermal storage is used in most homes today for water heating. PCMs are now being considered in a vari-

ety of building end uses in both passive and dynamic ways^[21,22] to increase the storage capacity and better integrate TES with existing structures and systems. As depicted in **Figure 1**, most passive TES applications incorporate PCMs into elements of the building envelope, where they provide additional thermal mass and are charged and discharged by diurnal changes in indoor and/or outdoor ambient temperature.^[21–25] Alternatively, dynamic applications typically integrate PCMs into building thermal equipment, such as air conditioning, domestic hot water, refrigeration, and space heating. In these applications, the PCM is charged and discharged directly by the thermal equipment (e.g., vapor compression system or electric resistance heat) to shift electric loads associated with each thermal end use.^[26–39] A common dynamic application is the use of campus-scale cold storage, using large chillers with ice or chilled water as the storage medium.^[40] While this is a cost-effective TES solution for campus systems and large commercial buildings, there is a need to integrate TES into smaller, modular building systems, as all residential buildings and most commercial building floor space are conditioned by smaller packaged air conditioners and heat pumps.^[3] Additive manufacturing can play a leading role in developing power-dense, high-efficiency compact PCM-integrated heat exchangers for such applications.

Beyond building applications, there is a valuable role TES can play in power systems and industrial processes. PCMs with phase change temperature greater than 200 °C have been studied for use as TES in large-scale concentrated solar power (CSP) plant applications,^[41,42] and in pumped heat electrical storage (PHES) plants, also known as Carnot batteries.^[43,44] For example, one study found that heat recovery utilizing PCM thermal energy storage resulted in 50–70% energy savings related to heating an industrial batch process for chemical manufacturing.^[45] There have also been studies exploring the use of PCM TES to transport waste heat from industrial sources to demand locations, such as hospitals.^[46]

Finally, certain electronic devices such as batteries, laser systems, or electric vehicle power electronics are highly transient and require pulse heat dissipation. A heat sink, or thermal

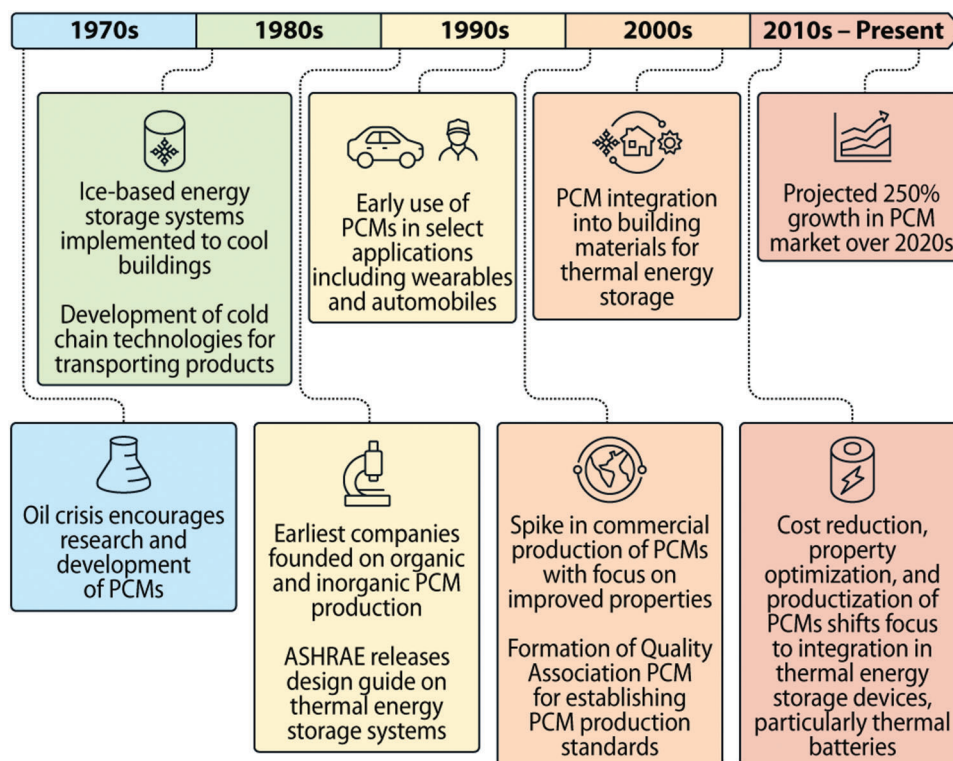


Figure 2. Graphical timeline of major events in the commercialization of PCM and PCM-integrated products.

management device, made of PCM can absorb large heat spikes while maintaining a constant temperature.^[47–52] This has been an active area of research for additive manufacturing, which excels at creating optimized extended surfaces for both traditional and PCM-based heat sinks.^[53,54] Also, many products must stay below a maximum allowable temperature, and PCM integration can extend how long such sensitive goods can stay viable for transport purposes. Examples in the literature include vaccines^[55,56] and temperature-sensitive foods.^[57,58] For thermal management at the personal level, PCMs have been used for thermotherapy/medical dressings^[59,60] as well as integration into textiles, sportswear, and bedding materials.^[61] Even vehicle systems can benefit from the use of PCMs for thermal management, allowing for buffering of peak thermal loads and for energy storage to improve cold start performance.^[62–64]

1.2. State of PCM Commercial Development

For a comprehensive review of the state of PCM-based products across multiple industries, see the recent work by Mehling et al.^[65] A graphical timeline of major advances in the commercial development and implementation of PCMs is outlined in Figure 2.

Several companies have developed PCMs with a range of chemistries. These companies, listed alphabetically, include (but are not limited to) Axiotherm,^[66] Boyd Corporation,^[67] Climator Sweden AB,^[68] Crodatherm,^[69] Eastex,^[70] ExxonMobil,^[71] Henkel Electronics Materials,^[72] Honeywell Electronic Materials,^[73] Insolcorp,^[74] Laird (DuPont),^[75] Microtek

Laboratories,^[76] Phase Change Solutions,^[77] Pluss Advanced Technologies,^[78] PureTemp,^[79] Rubitherm,^[80] Sasol Limited,^[81] Sunamp,^[82] and Tempered Entropy.^[83] While PCMs have an extensive history of application research, the transition to commercially available products has been somewhat limited. Reasons for the lack of industrial adoption include cost as well as limitations in thermal conductivity, latent heat, and charging/discharging, though these properties have experienced significant improvement over the last few decades. PCMs have traditionally been most successfully incorporated into thermal management devices for cooling electronics and in cold chain applications for temperature-controlled transportation of goods.

1.3. Additive Manufacturing and PCMs

The current state of the art of many latent heat TES systems involves large PCM-filled tanks with immersed heat exchanger tubing with diameters between ≈ 10 and 40 mm (Figure 3). However, recent advances in PCM materials with lower coefficients of expansion allow for smaller heat exchanger length scales with high surface area to volume ratios (length scales on the order of ≈ 1 –3 mm, see Figure 3). Additive manufacturing processes can be a powerful tool in this effort by enabling integrated complex fins or flow geometries not possible with traditional manufacturing. In addition, additive manufacturing facilitates the use of alternative heat exchanger construction materials, such as composite polymers comprising PCM or thermal conductivity-enhancing additives. One advantage of polymer materials is that they are less susceptible to

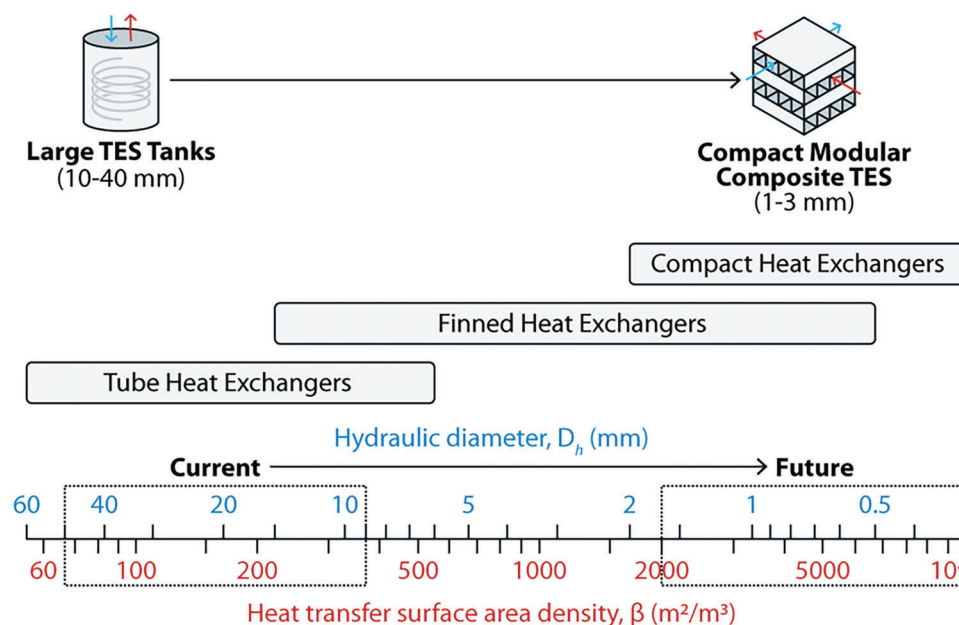


Figure 3. Vision for the future of TES heat exchangers with respect to magnitude of length scales and surface-area-to-volume ratios.

fouling, often resulting in a lower overall thermal resistance compared to traditional metal heat exchangers. Other advantages include lower required processing temperatures, lighter weight, and better cost-effectiveness.^[84]

Additive manufacturing can be a useful tool for TES devices because it can reduce the thermal resistance between the heat source/sink and the PCM. This is typically achieved in one of two ways—by increasing the effective thermal conductivity of the PCM with conductive scaffolding or by modifying the geometry to increase surface area and reduce heat transfer length scales. Geometries that are used for extended surfaces for conductive scaffolding are often complex to take full advantage of the capabilities of AM, as shown in **Figure 4**. These methods require some high thermal conductivity for the AM structure, usually metal or a polymer with thermally conductive additives.

Ideally, with the advent of additive manufacturing for TES heat exchangers, both the energy density and the power density can be co-optimized. As shown in **Figure 5**, AM allows for creation of ultra-compact high-surface-area-to-volume heat exchangers with advanced geometries that reduce overall weight while maximizing PCM volume and lowering resistances without necessarily requiring high thermal conductivity materials. AM can also enable thinner walls, further reducing thermal resistance.^[89] These geometry-enabled advancements are not practical with traditional manufacturing techniques.^[85] Current AM resolution, typically on the order of tens-of-microns, would likely not inhibit PCM phase transitions, which experience deterioration in latent heats and shifts in transition temperatures with confinement in the sub-micron volume regime.^[90–92] Recent research has shown the ability to directly 3D print functional composite

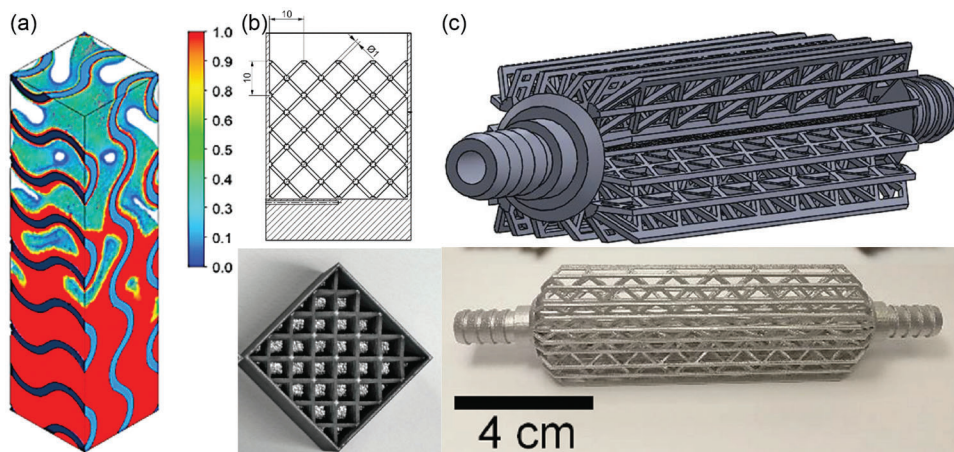


Figure 4. Complex conductive scaffolding structures made with additive manufacturing for heat transfer enhancement for PCMs in literature. a) Reproduced with permission.^[86] Copyright 2021 Elsevier Masson SAS. b) Reproduced with permission.^[87] Copyright 2020 Elsevier Masson SAS. c) Reproduced with permission.^[88] Copyright 2020, Elsevier.

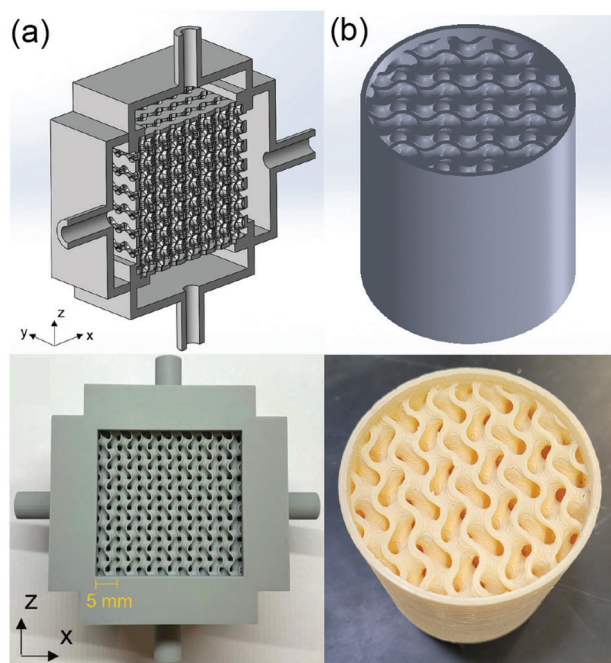


Figure 5. High surface area heat exchanger structures made with additive manufacturing for a) fluids, reproduced according to the terms of the CC BY license.^[99] Copyright 2022, the authors, published by Elsevier. and b) PCMs. Image of sample created during work published in ref [97].

filament with PCM directly incorporated into the structural wall material, which enables integration of the PCM storage and heat exchanger into one unified part, further reducing internal thermal resistances.^[93–98] This is discussed in depth later in this article in Section 2.2.

2. Advanced Materials and Additive Manufacturing for Phase Change TES

2.1. Review of Solid–Liquid PCMs

Solid–liquid PCMs are either organic, inorganic, or a eutectic (precise combination) of two organic or inorganic PCMs. Organic PCMs include paraffins,^[100] sugar alcohols,^[101] and fatty acids along with their alcohol and ester derivatives.^[102] Inorganic PCMs include salts,^[103] salt hydrates,^[104] salt–water solutions,^[105] and metals.^[106] Eutectics are typically combinations of two or more organic and/or inorganic PCMs with a single, minimum transition temperature.^[107,108] Inorganic PCMs generally have higher energy density/specific energy, higher thermal conductivity, lower cost (except for metals), and are naturally occurring (i.e., not petroleum or chemically derived). However, they are often corrosive, and tend to suffer from materials challenges such as supercooling^[109,110] and phase separation,^[110] which limit end-use reliability. Except for sugar alcohols, organic PCMs do not supercool. They are also usually single-component materials that are not prone to phase separation. However, organic PCMs are often petroleum derived, with sustainability issues, while limited bio-derived options are available but at greater costs. They typically have lower energy density/specific energy and are often

flammable. In general, organic PCMs are easier to incorporate into AM workflows because they are easier to microencapsulate, are stable at higher temperatures (which is required by some AM processes), are minimally corrosive to metals, and do not run a risk of phase separating during AM processing. Some advantages and disadvantages of organic versus inorganic PCMs are shown in Table 1. For many applications, the PCM transition temperature may be the first/most important PCM selection criteria as some PCM types do not have a wide range of available transition temperatures (Figure 6). Salt hydrates, for example, are typically not available above transition temperatures of 100 °C. Sugar alcohols are typically not available below transition temperatures of 80 °C.

Figure 7 depicts a comparison of the properties of different PCM types based on their material properties: latent heat by volume, latent heat by mass, cyclability, thermal conductivity, density, corrosiveness, nucleation ability (i.e., susceptibility to supercooling), and AM compatibility, or ease of integration of that PCM type into AM workflows. The comparison is relative, meaning the PCM type with the best performance for a given property is scored at 100% (exterior of circle), while the PCM type with the worst relative performance for that property is scored lower and toward the interior of the circle. For example, metallic PCMs have very high thermal conductivity, while most other PCMs have lower thermal conductivity. Metals as PCMs do not supercool, and have high density and thermal conductivity, and high cycle life. However, their latent heat is typically quite low. Organic PCMs (paraffins and non-paraffins) also have strong nucleating ability (do not supercool), long cycle life, and decent latent heat of fusion by mass. However, they have low thermal conductivity, density, and volumetric latent heat. Salt-based PCMs typically have relatively high latent heat (by mass and volume), density, and higher thermal conductivity than organic PCMs. However, they are corrosive, prone to supercooling, and can have poor cycle life due to phase separation. Organic PCMs (paraffins and non-paraffins) and metals/alloys are readily integrated into AM workflows. When processed via AM approaches, organic PCMs behave like many polymers that are already readily 3D printed. Similarly, methods to process metals/alloys via AM workflows are well developed. Salt-based PCMs, however, are difficult to microencapsulate and can be corrosive to metal components in AM processing equipment when used as is. Furthermore, many AM processes require high temperature, which can dry out salt-based PCMs with a water component (i.e., salt-hydrates, salt–water solutions), degrading them and changing their properties.

2.2. Processing of PCM Composites via Additive Manufacturing

All AM PCM components contain two key elements—the PCM itself and a matrix to contain and potentially enhance the properties of the PCM. The nature of the matrix material selected is directly related to the AM method with which it is processed. These matrix materials are used to develop 1) filaments, 2) resins, 3) inks, and 4) powders (shown schematically in Figure 8) that are then assembled in a layer-by-layer fashion using thermal, evaporative, or light-based processing. As the AM PCM field has evolved, so has the relationship between the matrix and the PCM.

Table 1. Advantages and disadvantages of organic and inorganic PCMs.

	Organic	Inorganic
PCM types	– Paraffins – Non-paraffins	– Metals/alloys – Salts – Salt-hydrates – Salt-water solutions
Advantages	– Not corrosive – Little to no supercooling – Chemically and thermally stable – Wide range of transition temperatures	– High energy density – Higher thermal conductivity – Lower cost – Nonflammable – Naturally occurring
Disadvantages	– Low energy density – Low thermal conductivity – Often flammable – Often petroleum based	– Prone to supercooling – Prone to phase separation – Often corrosive

In the most common efforts, a device is fabricated via AM and backfilled with a PCM. Increasingly, material and printing advancements have enabled direct incorporation of PCMs into the matrix material through physical blending, particle dispersion, coaxial extrusion, or chemical grafting, and in select cases, directly printing the PCM without a supporting matrix.

In the following sections (2.2.1–2.2.5), PCMs used in AM and prior research efforts where filament, resin, ink, or powder AM methods were used to directly incorporate PCM into a 3D-printed composite are reviewed. These are also summarized in Table 2. In Section 2.2.6, the compatibility of specific PCMs with AM processes is reviewed.

2.2.1. PCMs Used in Additive Manufacturing

The PCMs targeted in AM applications are largely organic in nature due to low corrosivity, high latent heats of transition, and limited supercooling while having thermal stability at elevated temperatures required for some AM processes. Most PCMs used

in AM applications exhibit solid-liquid transition behavior, which generally have higher latent heats compared to solid-solid transitions.

Commonly utilized PCMs for AM applications are 1) paraffins,^[88,118,126–128,131,134–139] 2) other alkanes,^[86,118,122,139–144] and 3) polyethylene glycols (PEGs).^[113,129,145] Other PCMs used include a sugar alcohol, fatty alcohol, fatty acid, and water.^[116,132,137,146,147] Commercialized PCMs are routinely utilized given the variety of transition temperatures and are available both in neat^[87,93,94,112,148–163] and microencapsulated forms.^[97,111,112] In an example of a metal PCM, researchers capitalized on the solid-state transition of a nickel titanium alloy from monoclinic Martensite configurations to a cubic Austenite lattice for AM metallic parts.^[124] Another case of metal PCMs used a printed aluminum-tin alloy where tin was the PCM given its comparatively low melting temperature.^[123] Other niche PCMs include a thiolated octadecane derivative and stearyl acrylates, which are derivatives of the fatty acid stearic acid intended for chemical grafting and polymerization.^[125,130,131,133]

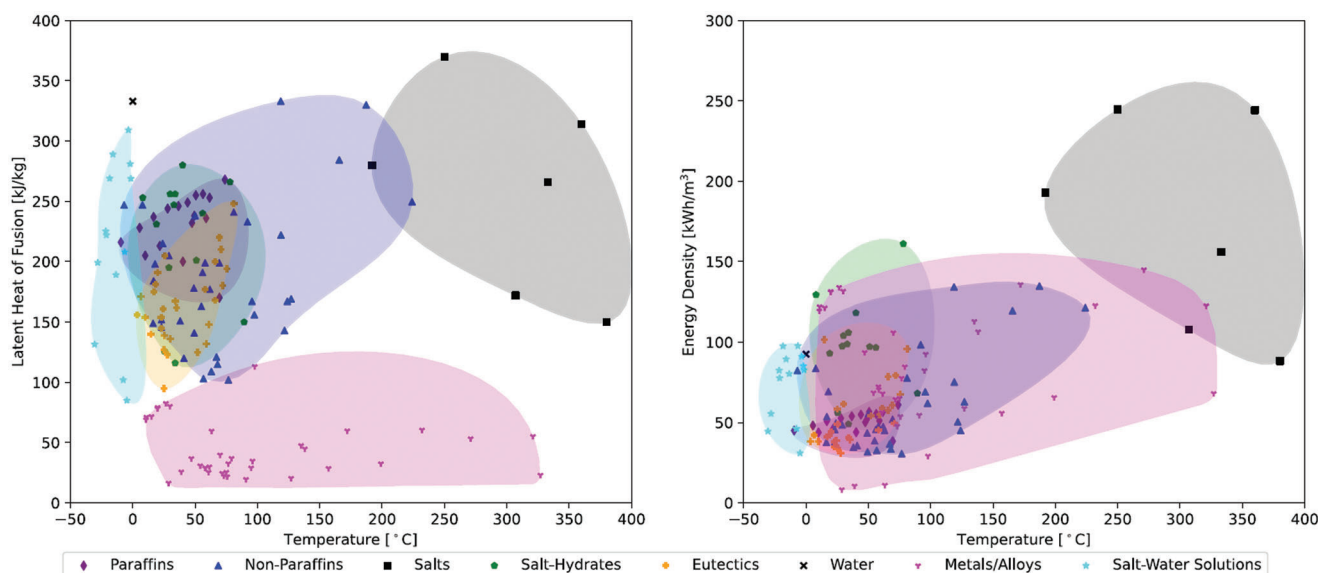


Figure 6. Latent heat of fusion and energy density versus phase transition temperature for the various PCM categories.

Table 2. Summary of research works incorporating PCMs directly into additively manufactured composites.

Author	Printing Method	Printing Material	PCM	PCM Encapsulation	PCM Loading	Size [L×W×H]	Transition Temperature [°C]	Latent Heat [J g ⁻¹]	Thermal conductivity [W m ⁻¹ K ⁻¹]
D. Rigotti, 2018 ^[111]	Filament	Thermoplastic polyurethane	MEPCM (Nextek 6D)	Powder blend	30-60 wt.%	n/a	0	70	n/a
P. Singh, 2022 ^[97]	Filament	Nylon	MEPCM (Nextek 6D)	Powder blend	30-40 wt.%	110 × 100 × 100 mm	6	46	0.14
R. Salgado-Pizarro, 2021 ^[112]	Filament	Polycaprolactone	Paraffin RT27	Melt blend	10-60 wt.%	35 × 12 × 2 mm	27	90	n/a
R. Salgado-Pizarro, 2021 ^[112]	Filament	Polycaprolactone	Micronal DS 5040	Solution blend	10-60 wt.%	n/a	22	40	n/a
T. Freeman, 2019 ^[93]	Filament	High density polyethylene	PureTemp PCM 42	Melt blend	20-60 wt.%	n/a	36	44	n/a
T. Freeman, 2021 ^[94]	Filament	High density polyethylene	PureTemp PCM 42	Melt blend	20-60 wt.%	25 × 25 × 4 mm	37.7	63.7	0.365
Z. Yang, 2022 ^[113]	Filament	Thermoplastic polyurethane	PEG 8000	Graft	50 mol%	200 × 200 × 0.4 mm	55	65	0.45
C. Wei, 2014 ^[114]	Ink	Paraffin wax	Paraffin	n/a	100 wt.%	n/a	n/a	n/a	n/a
H. Cui, 2022 ^[115]	Ink	Cement	MEPCM	Blend	10-20 wt.%	235 × 235 × 210 mm	n/a	n/a	0.551
J. Gasca-Tirado, 2019 ^[116]	Ink	Cement	Dodecanol	Suspension blend	3.68 wt.%	n/a	21.6	7.427	n/a
L. Hao, 2022 ^[117]	Ink	Cement	Paraffin	Suspension blend	6-25 wt.%	400 mm × 400 mm × 400 mm (hollow, 50 mm wall)	36.52	40.65	1.176
P. Wei, 2021 ^[118]	Ink	Elastic resin	Paraffin, eicosane, hexatriacontane	Suspension blend	41-67 wt.%	95×67×100 mm (hollow, 10 mm wall)	36 – 76	102.74	n/a
Q. Shahzad, 2021 ^[119]	Ink	Cement	Hentriacontane	Suspension blend	5-15 wt.%	~ meter scale	n/a	n/a	n/a
Y. Han, 2014 ^[120]	Ink	Paraffin wax	Paraffin	n/a	100 wt.%	0.05 × 0.05 × 0.05 mm	n/a	n/a	n/a
Y. Han, 2015 ^[121]	Ink	Paraffin wax	Paraffin	n/a	100 wt.%	n/a	n/a	n/a	n/a
Z. Yang, 2021 ^[122]	Ink	Graphene oxide/water	Octadecane	Coaxial extrusion	n/a	20 × 20 mm × 3 mm	27.5	190	1.67
C. Con-falonieri, 2021 ^[123]	Powder	Aluminum-Tin Alloy	Tin	Powder blend	21 vol%	40 × 5 × 9 mm	230	23.54	150 – 200
D. Sharar, 2020 ^[124]	Powder	Nickel-Titanium Alloy	NiTi Alloy	Powder blend	100 wt.%	25.4 × 25.4 × 3 mm	25	16.19	6 to 14
E. Wilts, 2020 ^[125]	Powder	Poly(stearyl acrylate)	Poly(stearyl acrylate)	Graft	100 wt.%	50 mm	42.8	82.4	n/a
M. Nofal, 2017 ^[126]	Powder	Graphite	Paraffin	Powder blend	80 wt.%	50 × 50 × 1.5 mm	56	163	0.7-1.86
M. Nofal, 2019 ^[127]	Powder	Graphite	Paraffin	Powder blend	80 wt.%	50 × 50 × 1.5 mm	55	150	0.83 – 0.92
M. Nofal, 2020 ^[128]	Powder	Graphite	Paraffin	Powder blend	80 wt.%	27 × 20 × 1.7 mm	55	155	0.85
C. Wang, 2022 ^[129]	Resin	Acrylic ester	PEG 400/PEG 1000	Melt blend	50 wt.%	5 × 5 × 35 mm	34.7	7.2	n/a

(Continued)

Table 2. (Continued).

Author	Printing Method	Printing Material	PCM	PCM Encapsulation	PCM Loading	Size [L×W×H]	Transition Temperature [°C]	Latent Heat [J g ⁻¹]	Thermal conductivity [W m ⁻¹ K ⁻¹]
J. Ma, 2021 ^[130]	Resin	Polymethylvinylsiloxanes	1-octadecanethiol	Graft	50 mol%	n/a	53.6	125.3	0.24
J. Ma, 2022 ^[131]	Resin	Polymethylvinylsiloxane	1-octadecanethiol (and other waxes)	Graft	50 mol%	n/a	80.4	195	1.56
P. Gogoi, 2020 ^[132]	Resin	Acrylamide/Alginate	Water	Sorption	n/a	30 × 30 × 60 mm	n/a	n/a	n/a
Y. Mao, 2018 ^[133]	Resin	Stearyl acrylate	Stearyl acrylate	Graft	25-50 mol%	40 × 40 × 2 mm	39.88	56.68	n/a

2.2.2. Filament Materials

Filament-based AM is one of the most widely used forms of AM because of its low equipment and material costs, limited post processing requirements, and diversity in availability of materials. Compared with other AM techniques, filament-based processes have lower layer resolution, weaker interlayer adhesion, and may require support structures for complex parts. Given the ease of filament accessibility and processing, incorporation of PCMs has taken several forms: 1) neat PCM blending, 2) dispersion of microencapsulated PCMs (MEPCMs), and 3) copolymerization. All filaments that encapsulate PCMs have been polymers like high density polyethylene (HDPE),^[93,94] polycaprolactone (PCL),^[112] thermoplastic polyurethanes (TPU),^[111,113] acrylonitrile-butadiene-styrene (ABS),^[155,163,164] and nylon.^[97]

For filaments incorporating neat PCM, encapsulation relies on solubility and uptake of the PCM into the filament material that remains upon extrusion and printing.^[93,94,112] Addition of com-

mercial PCMs to matrix materials has been done in the solution phase with PCL as well as mechanical mixing in the melt state prior to extrusion with HDPE.^[93,94,112] Using HDPE as a matrix, 40 wt.% of PCM was mixed in with good filament extrusion properties and a transition enthalpy of 64 J g⁻¹.^[94,95] Using PCL as a matrix, PCM additions up to 60 wt.% were made, but only 20 wt.% PCM filaments were printed, reducing the PCM transition enthalpy from 25 to 10 J g⁻¹ in the printed state.^[112]

MEPCMs have been incorporated into nylon, TPU, and PCL filament types. Methods of integration can be divided into (a) powder mixing and (b) solution dispersion. In two examples, MEPCMs were mixed with polymer powders and extruded directly into filament achieving MEPCM contents between 30 and 50 wt.% and exhibited melt enthalpies in the range of 46–70 J g⁻¹ for the printed parts.^[97,111] Using solution processes, PCL was dissolved in chloroform and 60 wt.% MEPCM was dispersed in the solution prior to drying and breaking up into pellets that had a melt enthalpy around 40 J g⁻¹.^[112]

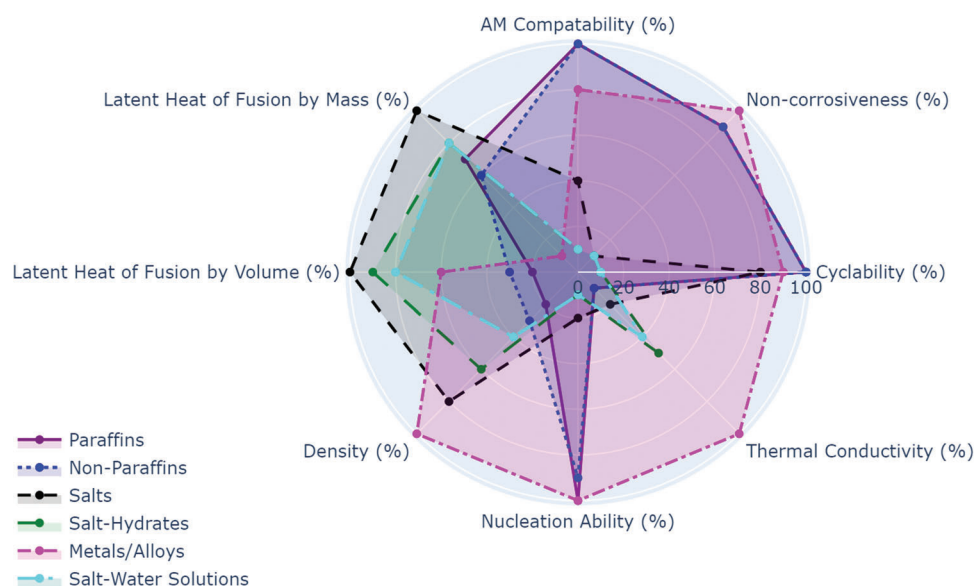


Figure 7. Comparison of PCM properties.

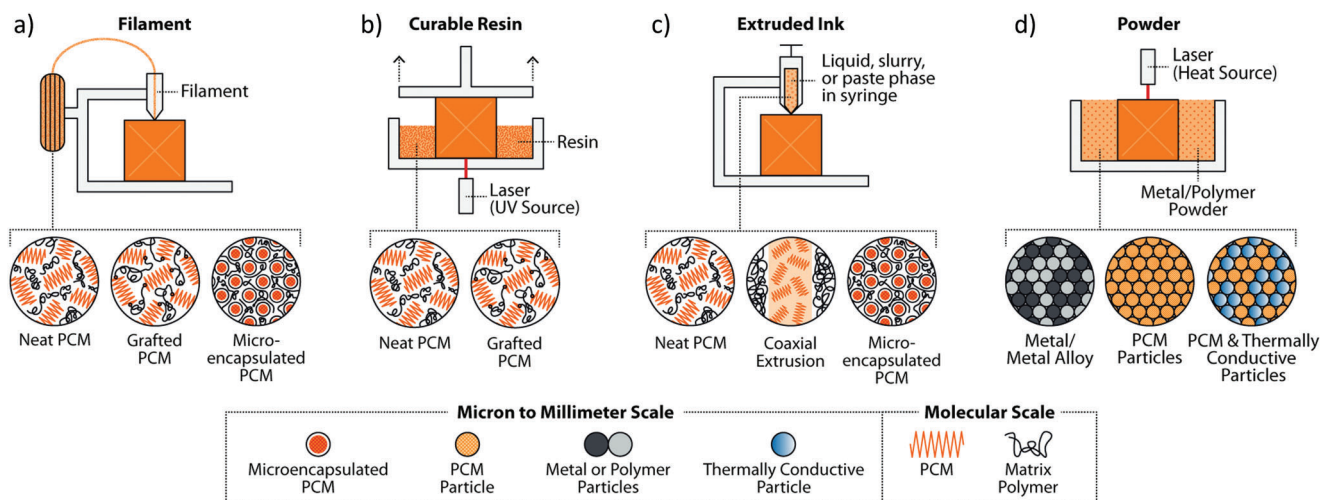


Figure 8. Graphical depiction of additive manufacturing methods used for PCMs; a) fused filament fabrication (FFF), also known as fused deposition modeling (FDM); b) stereolithography (SLA); and c) direct ink writing (DIW); d) selective laser sintering (SLS), also known as selective laser melting (SLM). Varying forms of AM materials with different methods to incorporate PCMs into the matrix.

Along with the use of MEPCMs to prevent PCM leakage from filaments, direct covalent bonding of the PCM with the filament matrix material has been shown to be an effective method to achieve TES in a printable material without leakage. Yang et al. capitalized on the hydroxyl end groups of PEG 8000 to copolymerize with an isocyanate-terminated TPU precursor to form domains of PEG that could melt and crystallize while tethered to the matrix materials, exhibiting a quasi-solid-solid phase transition with a transition enthalpy of 65 J g^{-1} , which did not diminish even after >2000 cycles.^[113]

2.2.3. Resin Materials

Resin-based AM relies on crosslinking photocurable chemical groups with a laser at each layer. This process, stereolithography (SLA), produces high-resolution objects at a relatively rapid rate. Compared with other AM techniques, resins can be costly and hazardous, with printed parts requiring extensive post-processing and supporting material that cannot be reused. The types of resin materials that have been combined with PCMs include: acrylamides,^[132] acrylic esters,^[129] siloxanes,^[130,131] and stearyl acrylates.^[133] An important limitation of resin approaches is that any material mixed into the resin must not optically interfere with the light source that initiates photocuring. As such, dark or highly opaque materials can be problematic.

When integrating PCMs into resin-based systems, researchers have blended neat PCM into resins as well as covalently linked PCMs into the resin matrix. With PCMs blended into resins, the PCM domains would ideally be locked into the crosslinked structure with curing. Wang et al. mixed two PEG PCMs into an acrylic ester resin and printed a simple rectangular prism ($5 \times 5 \times 35 \text{ mm}$) with layer resolutions of $50 \mu\text{m}$. At temperatures above PCM melting, PEG seeped to the surface of the print due to volume expansion of the PCM. The PEG seepage, coupled with a low latent heat, 7.2 J g^{-1} , indicates substantial durability issues with the composite.^[129] Gogoi et al. used water as the PCM absorbed into a

flexible, SLA-printed acrylamide-alginate hydrogel supported by PDMS at a size of $30 \times 30 \times 60 \text{ mm}$. In this example, water is supported through secondary intermolecular interactions and the flexibility of the hydrogel-PDMS system accommodates the volumetric changes with water during phase transitions.^[132]

A few efforts have used derivatives of solid-liquid PCMs that can be grafted to polymer backbones or directly polymerized into a brush configuration. Singly tethered ends of attached PCMs restrict mobility and inhibit leakage while leaving the other end free to undergo crystallization and melt events exhibiting a quasi-solid-solid phase transition. Ma et al. used thiolated derivatives of octadecane as a PCM with the ability to be grafted to siloxane and wax matrix materials and achieved transition enthalpies in the range of $25\text{--}125 \text{ J g}^{-1}$ with print sizes around $30 \times 30 \times 5 \text{ mm}$.^[130,131] In one of these siloxane-octadecane-wax materials, it was noted that the surface showed undulations with heating, indicating some effects of volume changing during a phase change, but this did not impact the transition behavior over 100 thermal cycles. Utilizing a macromonomer approach, Mao et al. used an acrylate derivative of stearic acid along with an acrylamide monomer and crosslinker. These were copolymerized during the printing process to form a polymer network that had a transition enthalpy around 70 J g^{-1} .^[133]

2.2.4. Ink Materials (Direct Ink Writing)

Ink-based materials in the AM context refer to liquids or slurries that are directly extruded in a layer-by-layer fashion in a process called direct ink writing (DIW). As the ink is deposited, solidification of each layer can occur through thermal curing, photocuring, solvent evaporation, or reactions with the solvent.^[165] Because ink-based AM can accommodate numerous curing processes and can be performed at low temperatures, there is large diversity in material selection, which is advantageous over other, more restrictive techniques. A notable disadvantage of ink-based AM includes the high degree of control required over the

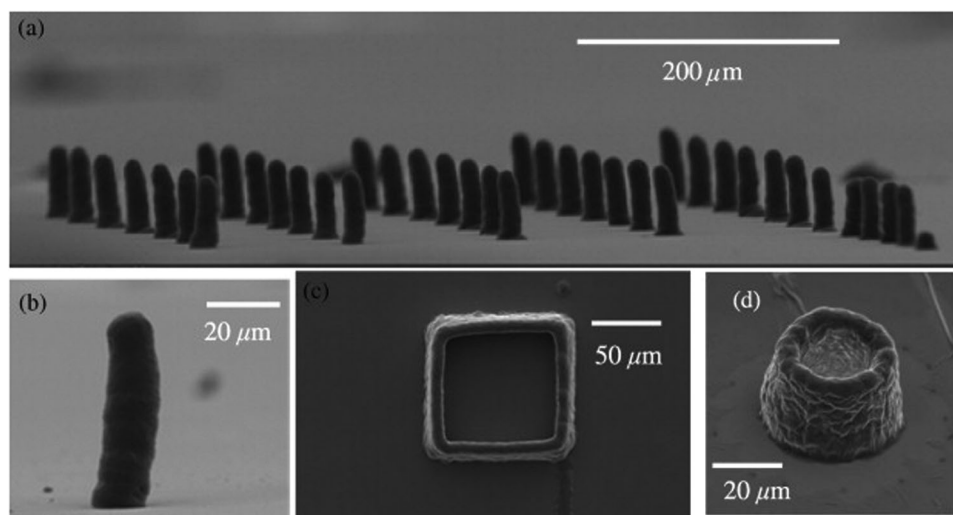


Figure 9. Micro-structures printed from EHD printing process. a) Micro-pillar array. b) Close view of a single pillar. c,d) Square and circular tube with thin walls. Reproduced with permission.^[120] Copyright 2014, Elsevier.

rheological properties to achieve high resolution and cohesive prints. Matrix materials that have been targeted for DIW of PCM composites include siloxane,^[131] cementitious composites,^[115–117,166] graphene oxide,^[122] and a commercial UV curable resin.^[167] The thiolated octadecane grafted to a siloxane matrix material developed by Ma et al. discussed in the previous section was demonstrated to be able to be processed via DIW utilizing a high temperature UV curing step after printing.^[131]

In an example of coaxial DIW, Yang et al. demonstrated a thermally conductive graphene oxide ink printed as a sheath around a pulsed octadecane core that demonstrated a composite transition enthalpy and thermal conductivity of 190 J g^{-1} and $1.67 \text{ W m}^{-1} \text{ K}^{-1}$, respectively.^[122] The pulsations of the PCM core created isolated “beans” within the graphene oxide sheath that mitigated leakage in the event of localized damage to the print avoiding total leakage of the PCM. Using a similar process of creating localized regions of PCM within the matrix, Wei et al. created small waxy beads using emulsion processes and suspended the beads in a commercial resin typically used for SLA printing.^[118] Because the ink could be extruded and each layer UV cured at temperatures below the 60°C melting point of the paraffin PCM, the matrix was cured around the PCM beads in the solid state and prevented leakage from the print to achieve a transition enthalpy of 102 J g^{-1} .

An innovative development in ink-based AM utilizes electrostatic potentials to deposit $<10 \mu\text{m}$ diameter minuscule droplets of paraffin on a substrate with extremely high resolution in a process deemed electrohydrodynamic (EHD) 3D printing (shown in Figure 9). While not focused on TES and PCM applications, the demonstration of high-resolution methods of paraffin ink printing has potential for advanced PCM composite fabrication with spatially selective deposition of PCMs at resolutions $<10 \mu\text{m}$.^[114,120,121]

An emerging field of DIW TES composites is focused on PCM incorporation into cementitious materials for building-scale thermal management.^[115–117,166] In these materials, PCMs are dispersed in cement slurries that are then deposited in a layer-by-

layer fashion. The highest transition enthalpy achieved in an AM cement-PCM composite was 40 J g^{-1} using a neat paraffin PCM.^[117]

2.2.5. Powder Materials

Fabrication of AM structures via powder processing utilizes a laser to selectively melt or sinter particles into a cohesive structure. These powder-based processes are known in the AM community as selective laser sintering (SLS), powder bed fusion, and direct laser metal sintering. Powder-based AM can yield high-resolution parts with high interlayer strength and moveable components without the need for printed supports. However, powder-based AM processes can be slower than other AM techniques, and the required hardware can be expensive.

The materials space of powder-based AM is largely dominated by metal powders. Metallic materials for AM offer high thermal conductivities but often at the expense of cost, corrosion resistance, and weight. In AM PCM devices, aluminum-silicon alloys, commonly containing magnesium, have been the most widely used due to their high strength-to-weight ratio, corrosion resistance, and ease of powder processing through laser sintering. The most common alloy is AlSi10Mg,^[86–88,134,135,137,141,144–146,150,152,160–162] but other aluminum-silicon formulations^[142,143,148,154] and aluminum alloys^[123,136,138,149,151,157–159,168] have also been used. Other metals printed for PCM applications have included copper alloys,^[139,159] steel alloys,^[156,159] nickel alloys,^[124] and titanium alloys.^[159] These efforts produce metal structures that are generally fabricated to be porous for backfilling with PCM and focus on optimizing geometries that encourage rapid charging/discharging of the PCM.^[86–88,134–164,168] The backfilling of metallic AM devices with bulk PCM capitalizes on high thermal conductivities, leak-resistant structures, and considerable PCM volumes for high energy capacity. A few AM metallic structures have sought to use the metal as the PCM. Sharar et al. examined

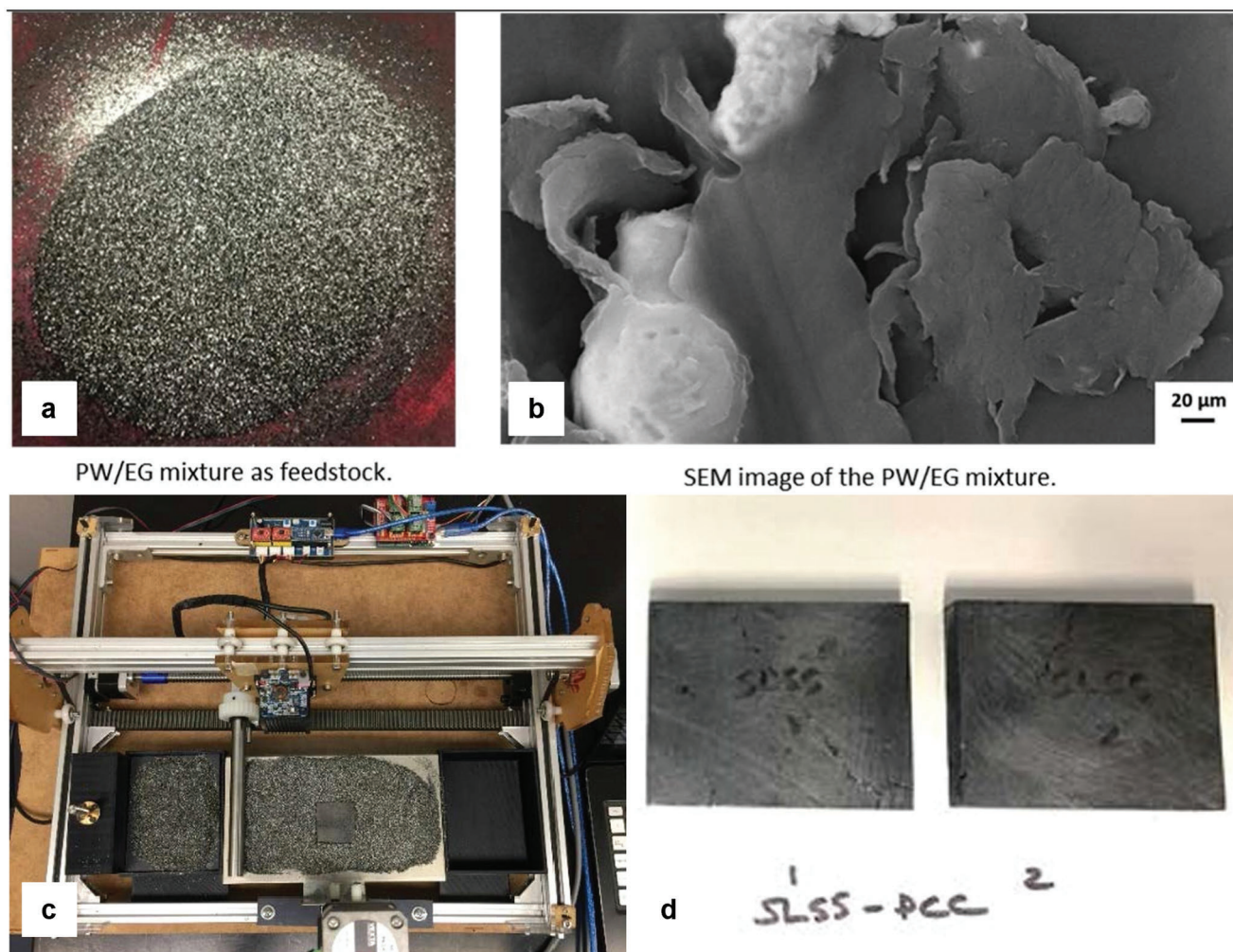


Figure 10. a) Powder mixture of paraffin wax (PW) and expanded graphite (EG). b) Scanning electron microscopy images of the PW and EG powder mixture. c) Powder bed AM setup for the laser sintering process. d) AM PW/EG composites fabricated from laser powder sintering. Reproduced with permission.^[128] Copyright 2020, Elsevier.

the solid-solid phase transition of a nickel–titanium alloy, achieving a transition enthalpy of 16 J g^{-1} at 25°C .^[124] In a solid-liquid transition of a metal PCM, Confalonieri et al. capitalized on the low miscibility of tin within aluminum and the lower melting temperature of tin to achieve a transition enthalpy of 24 J g^{-1} at 230°C .^[123] Some special cases of powder-based PCM composites exist where organic PCMs act as the binder during the powder fusion process. In a series of studies carried out by Nofal, et al., paraffin was used as both the PCM and the binder in combination with expanded graphite (EG) for laser sintering into multilayer composites in a variety of shapes that were 80% paraffin by weight and exhibited transition enthalpy of 155 J g^{-1} and thermal conductivity of $0.85 \text{ W m}^{-1} \text{ K}^{-1}$.^[126–128] Using laser sintering methods for EG-PCM composite fabrication (shown in **Figure 10**) circumvented traditionally lengthy and wasteful processing methods, such as press-soak manufacturing, that rely on PCM infiltration into EG powder prior to compaction.^[49,169] In another organic powder-based AM precursor, the derivative of the PCM stearic acid, stearyl acrylate, was polymerized via both suspension and solution polymerization to yield particles of variable

size that would allow for adequate coalescence during the powder fusion process. The parts fabricated using powdered poly(stearyl acrylate) exhibited a transition enthalpy of 82 J g^{-1} .^[125]

2.2.6. PCM Compatibility with Additive Manufacturing Processes

With the scope of work available in the literature, regarding inclusion of PCMs into AM processing, some broad conclusions about PCM compatibility across various AM techniques and promising future directions can be drawn.

MEPCMs have been well incorporated into filament-based and ink-based materials as a good approach towards TES composites.^[97,111,112,115] Although filament AM requires high temperatures, many MEPCM shells are stable at these temperatures.^[97] Mechanical damage to MEPCMs can occur during filament compounding leading to shell breakage, but this can be mitigated by using polymer powders over pellets.^[97] DIW processes are also a good option for MEPCMs, where the solid particles can be dispersed in a flowable matrix that

can be deposited into a layer prior to curing. With ink-based materials, there is reduced likelihood of mechanical damage from dispersal and ample opportunity to adjust viscosity. With several curing mechanisms for ink-based materials, MEPCMs could inhibit light transmittance in photocuring or deteriorate in the presence of some solvents for evaporative curing.^[112] Much like DIW processes, MEPCMs have potential in resin-based systems if they can stay in suspension, do not inhibit photocuring, or severely affect the viscosity. In polymer powder-based processing, MEPCMs would need to withstand the laser intensity without incurring damage.

Neat PCMs dispersed or blended into another material can be difficult to process due to leakage issues in printed structures. High temperatures required for filament processing can cause liquid PCM to escape, impacting PCM retention.^[112] Resin-based systems can bypass temperature issues but may be impacted by immiscibility/suspension of the PCM in the resin vat or interference with crosslinking chemistries that can affect PCM distribution in consecutive layers. There is also still the possibility of leakage, even from a cross-linked material due to volume changes by the PCM during transitions.^[129] With ink-based AM, low processing temperatures and direct material extrusion can allow solid-state PCM to be extruded and cured, enabling larger domains of PCM to exist within a device.^[118,122] Additionally, inks can be coaxially extruded, allowing neat PCM to be encased within an exterior matrix material as a means of encapsulation.^[122] Powder-based processing with neat PCMs is limiting and would likely require a porous matrix to absorb PCM during processing and would likely suffer from leakage or deterioration during long-term cycling.^[126–128]

Grafted PCMs exhibit quasi-solid–solid transitions due to the addition of covalent linkages with non-PCM species or sufficiently branched molecular architecture.^[113,125,130,131,133] Bypassing the leakage issues of traditional solid–liquid PCMs, grafted PCMs have been incorporated into filaments, resins, and powders, with opportunity for use in ink-based processes.^[113,125,130,131,133] Grafted PCMs have the greatest potential to behave like traditional thermoplastics or crosslinked resins due to the freedom of chemical selection and are likely the most easily adapted into any AM process.

Bulk PCMs are typically backfilled into AM-produced liquid-tight scaffolds with extensive work in metal powder-based printing for high-resolution, high-thermal conductivity devices.^[147,151,162,163] Other AM techniques, like resin-based systems can also produce liquid-tight parts, with possible addition of high thermal conductivity fillers to supplement thermal properties. Some filament-based parts can be made liquid-tight, but generally not as well as other AM techniques.

2.3. Thermal Conductivity Enhancement in PCM Additive Manufacturing

Base polymeric materials used in AM often suffer from low thermal conductivity as well. Additives, thereby, are often incorporated to increase the heat transfer rate of the composite material. Additives utilized for thermal conductivity enhancement are typically materials with inherently high thermal conductivity. Of note, thermal conductivities of additives and their effects on TES

devices can be highly variable depending on factors such as particle size, filler volume, crystallinity, and orientation within the composite.^[170] Shemelya et al. showed that anisotropic thermal conductivity for 3D-printed structures is related to the print direction and filler morphology, meaning that thermal conductivity can be controlled through a combination of print raster direction and material design.^[171] Therefore, further optimization techniques for AM components can be employed to improve the thermal conductivity.

During filament printing, anisotropy of the filament occurs due to the differing degrees of interdiffusion in the in-plane and out-of-plane directions of the filament and incongruent layering between the filament.^[172] Studies primarily focus on the mechanical properties of AM components; however anisotropic characteristics of the AM components can significantly vary along the print direction.^[173] Anisotropy occurs due a discontinuity of material across the print layers. Control over anisotropy can be achieved by modifying the polymer matrix and adjusting the print parameters.^[174,175] The particles or fillers used to enhance thermal conductivity provide conductive pathways within the raster direction of the composite when they are aligned in the proper orientation.

Improving heat transfer in AM objects can be controlled before, during, and after manufacturing by utilizing an appropriate system design for the AM component. Three parameters can be tailored to improve the heat transfer of AM components: 1) material selection and incorporation, 2) fabrication parameter optimization, and 3) post-processing techniques to improve thermal conductivity. Selection of the appropriate TC-enhancing additive, matrix material, and mixing ratio is crucial for good heat transfer and cohesive prints. During AM fabrication, an optimized design of the 3D model is generated, printing directions are chosen, and other printing parameters are determined to increase the adhesion between the printing bed and printed parts. The fabrication step also includes tailoring the surface roughness of the AM components based on printing conditions and slicing strategy to reduce the staircase effect,^[176,177] and controlling local thermal conditions through microstructure (i.e., grain nucleation in metal AM components).^[178,179] The post-processing treatment includes mechanical polishing and thermal or vapor annealing that can be directly applied to the surface of the printed part,^[180] which can improve thermal contact with other components, promoting better effective thermal conductivity and overall heat transfer.

Thermal conductivity in an AM component is generally lower than its parent material with the reduction most pronounced in the build direction.^[181–183] The reduction in the thermal conductivity and the inherent anisotropy of AM components is due, in part, to incomplete fusing and substantial thermal contact resistance between the layers.^[181] Certain AM processes lend to better interlayer adhesion and mitigate reductions in thermal conductivity.

2.3.1. Material Additives for High Thermal Conductivity

Materials development can be utilized with fundamental design principles for enhanced thermal conductivity. These composites often consist of a base polymer and a thermally conductive filler (e.g., graphene, boron nitride, aluminum oxide).^[184] For a

thorough review of material parameters, the reader is referred to Almuallim et al.^[179] The thermal conductivity of the base polymers is often on the order of 0.04–0.36 W m⁻¹ K⁻¹.^[185] High thermal conductivity fillers, like carbon derivatives, metals, or ceramics are added to polymer-based composites to enhance the overall thermal conductivity of the composite.

Additives based on carbon derivatives are available in a variety of allotropes and commonly include carbon fibers and nanotubes, graphene, and carbon black and have been widely studied for their effects on improving the thermal conductivity of polymer composites, as well as mechanical properties.^[186–188] Carbon-based additives have advantages of high thermal conductivity, chemical stability, and low density ($\approx 2 \text{ g cm}^{-3}$).^[189,190] Thermal conductivity measured at ambient conditions (25 °C) of expanded graphite (EG) can range from 4–400 W m⁻¹ K⁻¹,^[191] carbon nanotubes (CNT) 2000–6000 W m⁻¹ K⁻¹,^[192] and carbon black (CB) ≈ 30 –170 W m⁻¹ K⁻¹.^[193] In carbon-based additives, geometry (i.e., how the carbon additive is oriented within the material) plays a role in the improvement of thermal conductivity. Alignment of conductive fillers or fibers in the print direction can assist in the direction of heat flow through the printed material. Liao et al. showed that as the carbon fiber content increased above 4 wt.%, the thermal conductivity rapidly increased from 0.221 to 0.835 W m⁻¹ K⁻¹, which may be due to carbon fibers in the matrix establishing a continuous heat channel to efficiently enhance the conductivity, resulting from the preferential alignment of the carbon fibers in the print direction.^[194] Wang et al. utilized CNTs to prepare a composite aerogel, via melt deposition of the initial hydrogel, that exhibited anisotropic thermal conductivity of 0.025 W m⁻¹ K⁻¹ in the axial direction and 0.302 W m⁻¹ K⁻¹ in the radial direction as the CNT filler ratio increased.^[195] Shemelya et al. showed key thermal conductivity results for ABS/graphite composites, where the thermal conductivity was measured in the z-plane at 0.25 W m⁻¹ K⁻¹ and in the x-y plane at 0.37 W m⁻¹ K⁻¹.^[171] It was noted that the ABS/graphite composite exhibited noticeable thermal anisotropy due to the as-printed alignment of the graphite flakes.

Ceramic fillers such as boron nitride, silicon carbide, aluminum nitride, and silicon carbide are used for their high thermal conductivity, low coefficient of thermal expansion, and electrical resistance.^[179] Among these fillers, aluminum nitride has the highest thermal conductivity (140–180 W m⁻¹ K⁻¹), followed by silicon carbide (120 W m⁻¹ K⁻¹),^[196] and boron nitride (30 W m⁻¹ K⁻¹).^[197] Other niche additives include metal–organic frameworks (MOFs), titanium dioxide foam, and nickel foam.^[198] Li et al. investigated the particle size of hexagonal boron nitride (h-BN) with a fixed filler content on composite properties of FFF printed components using isotactic polypropylene.^[199] Their results show that the particle size of the h-BN filler had a significant effect on the thermal conductivity, with composites exhibiting 2.02 W m⁻¹ K⁻¹ parallel to the print direction. They concluded that the heat conduction formed by the aligned h-BN particles contributed to the improved thermal conductivity along the print direction. Sonsalla et al. tested Zn-SiC (zinc-silicon carbide) composites and determined that the addition of Zn-SiC improved thermal conductivity from 0.20 W m⁻¹ K⁻¹ in the horizontal direction for the ABS control to 0.42 W m⁻¹ K⁻¹, with further enhancement of 0.60 W m⁻¹ K⁻¹ in the vertical direction due to alignment with the filament print direction.^[200]

Ongoing research and development studies indicate that the challenges of improving the thermal conductivity of PCMs focus on the aspects of clarifying the phonon scattering mechanism in PCMs, increasing the number of thermal conductivity chains, and broadening the thermal transmission channels.^[198] Furthermore, increasing the thermal conductivity of AM composites with additives that have higher thermal conductivity than the parent matrix is affected by size, shape, and filler amount, which can play a role in the phonon transport properties of the matrix.^[179] As previously discussed, the introduction of additives reduces the thermal storage capacity (i.e., latent heat); therefore, it is important to consider not only additives but also relative amounts when improving thermal conductivity.

2.3.2. Fabrication Parameters for Improved Thermal Conductivity

Outside of materials development, improving thermal conductivity in AM components (to approach thermal conductivity of the pure material) can be done through optimization of printing parameters. Printed geometries such as lattice structure geometries can be utilized to increase the thermal conductivity in an AM component.^[135] Many studies report on the print parameters (e.g., layer height, fill density, print speed, nozzle diameter) for improvements in mechanical properties, but few report on the printer settings for maximized thermal conductivity of 3D-printed components. In optimizing the printing parameter, a study by Sonsalla et al. determined that the thermal conductivity of a 3D-printed structure was directly affected by fill density, layer height, and print speed. Nozzle diameter was found to be negligible in its effect on thermal conductivity.^[200] An increase in the thermal conductivity was observed with an increase in the fill density. Additionally, the authors demonstrated optimal layer height of 0.4 mm and fill density of 100% to achieve the highest thermal conductivity for a 3D-printed object fabricated with a homogenous material printed in a horizontal orientation. For metal powder printing, laser power output has a large influence on thermal conductivity of the final part. Particle size distribution and powder compaction will affect thermal dissipation during sintering, which with a low-power laser, will lead to incomplete consolidation (higher porosity) and reduced thermal conductivity of the printed object.^[201,202]

In another study by Elkholy et al., the effect of layer height and raster width and results showed that increasing the layer height and width causes a deterioration in the thermal conductivity up to 65% when compared to the pure polymer.^[203] The authors also found that decreasing the fill ratio decreases the thermal conductivity and that adding a carbon fiber filler improved the thermal conductivity in the z and y directions by 26% and 162%, respectively.^[203] They determined that the effect of raster width was more significant than the layer height in reducing the thermal conductivity in all directions due to porosity generation.^[203]

While optimization of printing parameters is important, this approach can be limited due to the process parameters and can negatively impact the production capacity (i.e., lower extrusion speeds reduce the output).

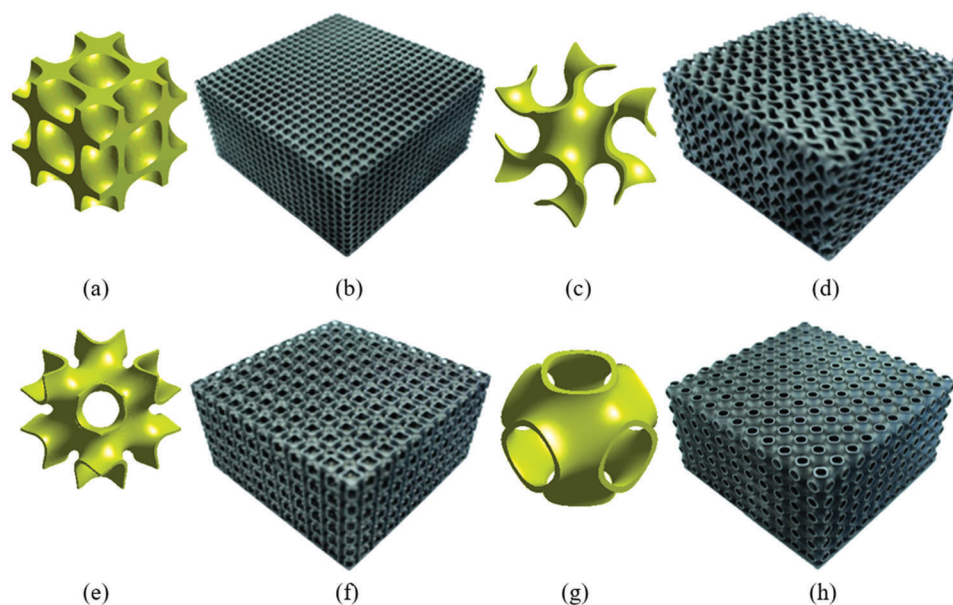


Figure 11. CAD models for unit cells for a) Diamond, c) Gyroid, e) IWP, g) Primitive; and 3D Printed TPMS lattices for b) Diamond, d) Gyroid, f) IWP, and h) Primitive. Reproduced with permission.^[152] Copyright 2021, Elsevier.

2.3.3. Postprocessing for Improved Thermal Conductivity

Postprocessing of AM components has been shown to improve the thermal conductivity in extruded components via thermal annealing.^[183] In traditional metal manufacturing, thermal annealing is used to improve the grain structure and contact resistance.^[204] The thermal contact resistance that exists between the layers of the AM components is often difficult to show due to difficulties in experimental measurements and is not reported widely in the literature.^[181] Thermal contact resistance occurs in the z direction due to the interfaces between the layers and may impede the heat flow and mechanical strength when compared to the x direction.^[181,205] Prajapati et al. showed that strong interfacial thermal contact resistance in the build direction was the cause of anisotropy, and that air gaps in the microstructure cause a reduction in the thermal conductivity in the measured raster direction.^[181] The authors showed significant improvements in the thermal conductivity when the polymers, ABS and ULTEM thermoplastic materials, were annealed at 135 °C for 96 h.^[181]

Vapor finishing techniques are used to smooth outer surfaces of the printed parts and serve to help lower thermal contact resistance with adjacent components at the surface; however, it is unclear if there is a significant impact on the interior geometry and thus the effective thermal conductivity of the printed parts.^[203]

2.4. Advanced Manufacturing for Integrating PCMs into Devices

2.4.1. 3D-Printed Scaffolds for Thermal Batteries

The most common method of combining PCMs with advanced manufacturing is by generating scaffolds that are then backfilled

with the PCM. By utilizing high thermal conductivity materials to build the scaffolds, these structures are capable of generating thermal batteries that can more quickly be charged and discharged relative to encapsulated PCMs alone. Furthermore, the ability to utilize computer-aided design (CAD) software to produce intricate scaffolds with high surface area allows for optimized advanced manufacturing. For instance, Qureshi et al. utilized aluminum AlSi10Mg gas-atomized powder to 3D print metal scaffolds via laser powder bed fusion, which produced intricate structures with high surface area and thermal conductivity. CAD drawings of these different structures are shown in **Figure 11**. With 85–90% porosity, the structures were then backfilled with molten paraffin wax PCM to generate thermal batteries.^[152]

Similarly, Guo et al. developed metal-printed scaffolds from AlSi10Mg through selective laser melting and backfilled them with n-tetradecane, resulting in a PCM thermal energy storage device with 13 times greater thermal conductivity than the pure PCM for faster charging/discharging.^[206] Yazdani et al. 3D printed an aluminum–silicon alloy (AlSi10Mg) grid as an extended surface on a flat plate heat exchanger using a selective laser melting process. The heat exchanger was tested with different grid designs, charging/discharging paraffin wax and myristic acid. The proposed application was cooling high-power 5G electronics and using the waste heat for other applications such as space heating.^[137] A parametric 2D FEM simulation was used for the design of an additively manufactured thermal storage device. Several designs were printed out of AlSi10Mg using direct laser metal sintering (DLMS) and tested in a coolant loop with paraffin wax as the PCM.^[88] Additionally, a maraging steel two-phase heat exchanger was manufactured using DLMS by Kabir et al. that was then backfilled with a commercial bio-based PCM, PureTemp25. The heat exchanger was designed for use as a thermal capacitor or thermal storage in spacecraft thermal management systems,

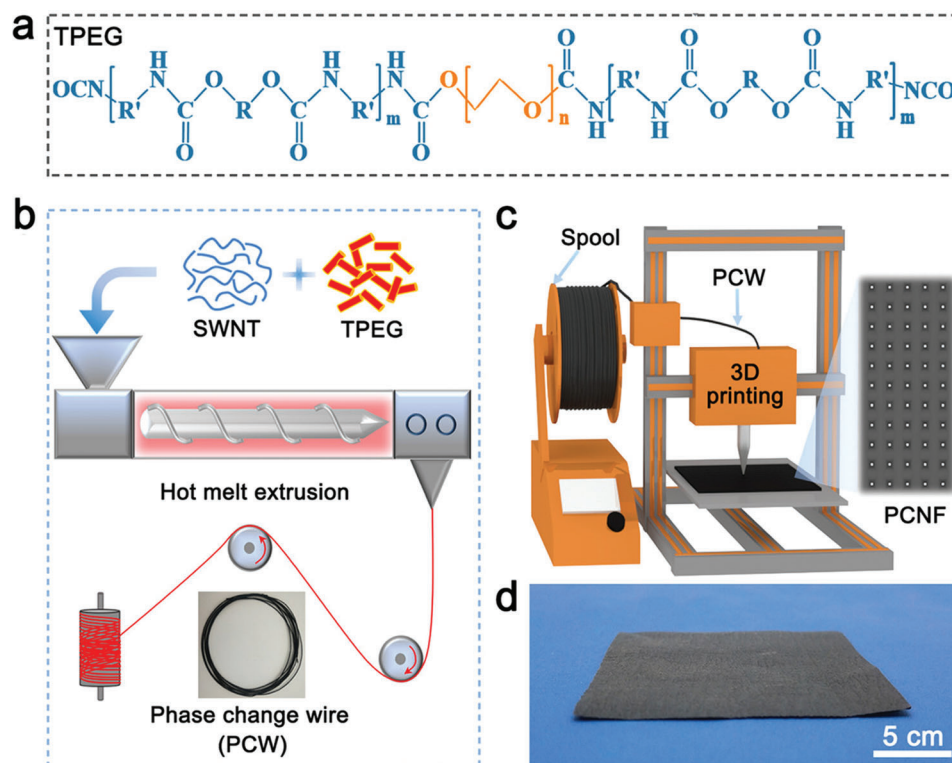


Figure 12. Fabrication procedure of 3D-printed phase change non-woven fabric (PCNF). a) Schematic representation for the preparation process of TPEG. Schematic illustrations for b) melt spinning of PCW and c) 3D printing of PCNF. d) Photograph of a large-scale PCNF. Reproduced with permission.^[113] Copyright 2022, American Chemical Society.

using methanol driven by capillary pumping pressure within the wick structure.^[156]

2.4.2. PCM-Integrated Materials for Wearable Technology

One unique application space for advanced manufactured PCM devices that has been studied is wearable technology. For example, acrylamide-alginate hydrogels developed by Gogoi et al. were designed as a wearable technology intended for cooling athletes during training and competitions. These composite prints possess a high level of wear resistance and mechanical stability suitable for this application. This technology is being explored for commercial use under the name CoolPak Hydrogel.^[132] In another study for wearable technology that incorporates PCMs, octadecane-grafted siloxane composites developed by Ma et al. deposited PCM on carbon fiber cloth, enabling temperature regulation for the wearer.^[130] They demonstrated the cooling potential with surface skin temperature reductions from 46 to 38 °C over 20 min. Crosslinked stearyl acrylate gel developed by Mao et al. was printed on fabrics such as cotton and demonstrated a prolonged equilibration time with the hot/cold stage indicating thermal regulation capabilities. The authors propose that this PCM-integrated gel on fabric may be applicable to the wearable technology market.^[133] PEG-TPU copolymer created by Yang et al. utilized single-walled carbon nanotubes (SWNT) to facilitate heat transfer to solid-solid PCM that was 3D printed into flexible sheets for cooling fabrics. At 1 wt.% SWNT, these sheets

possessed a thermal conductivity of $0.52 \text{ W m}^{-1} \text{ K}^{-1}$ and demonstrated rapid thermal regulation from 30 to 60°C and back to 30°C in just 2 min with latent heat of 65 J g^{-1} . The fabrication procedure of these sheets and some of their chemical characteristics are shown in **Figure 12**.^[113]

3. Perspectives on the Potential and Future of PCM Additive Manufacturing

As discussed in the introduction, there are many potential applications of PCMs—ranging from heating and cooling equipment to thermal management of electronic devices. Each application has its own constraints, which will impact how the PCM should be integrated. In devices used for heating and cooling, PCM TES allows for load shifting and peak shaving, reducing stress on the electrical grid and facilitating renewables integration. As previously mentioned in Section 1.1, including PCMs into various systems can result in one or more benefits, including energy efficiency (EE), emissions reduction (ER), load shaving/shifting (LS/S), and thermal management (TM), where TM is the ability of the PCM to prevent operation outside a system's safe or comfortable operating temperature range. **Figure 13** lists common applications of TES in the temperature range between -50 and 400 °C, as well as a measure of the potential of PCMs to provide EE, ER, LS/S, and TM benefits for each of the listed applications. For example, incorporating PCMs in building air-conditioning systems provides a LS/S benefit and could potentially provide EE and ER benefits depending on system

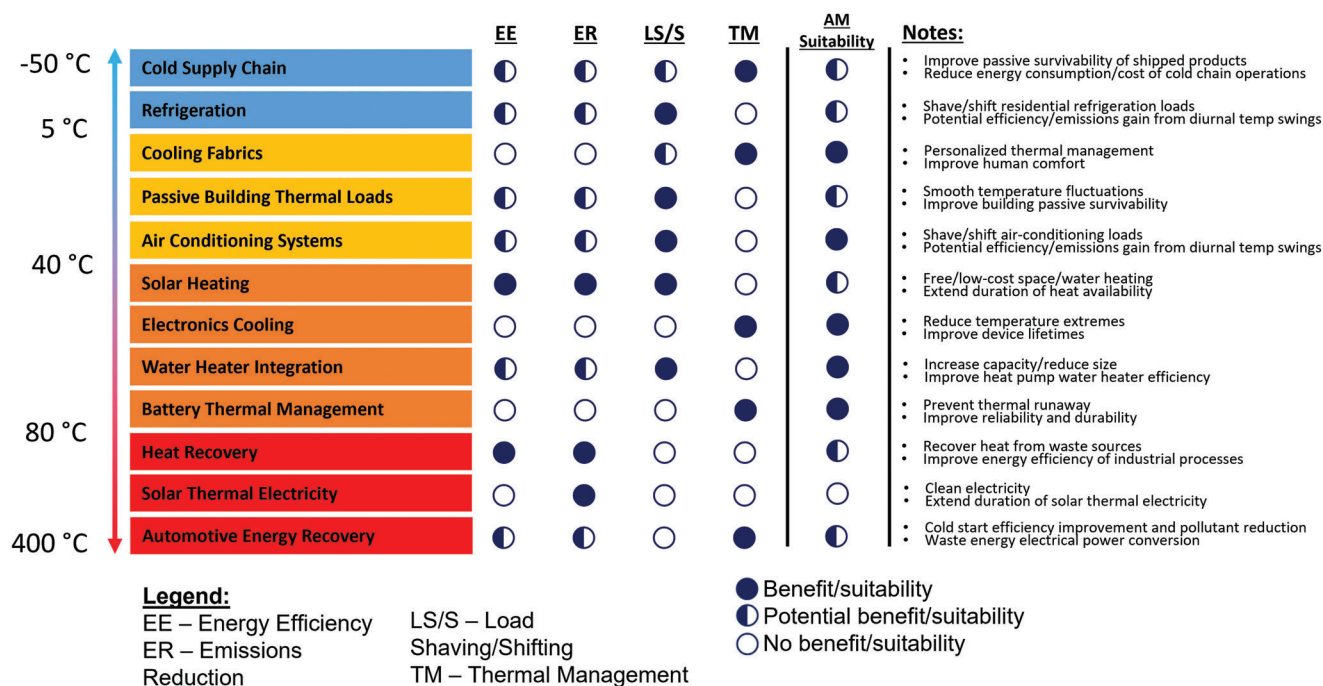


Figure 13. Common applications for PCMs across the temperature range of interest, the ability of PCMs to provide benefits to said applications, and the suitability of AM to impact said applications.

configuration and climate,^[3] but would not provide a TM benefit. Incorporating PCMs into textiles or cooling fabrics worn by humans provides comfort TM and potentially LS/S building loads if worn by enough occupants but is unlikely to provide EE or ER benefits. Figure 13 also classifies each application based on the suitability of additive manufacturing (AM Suitability) to have an impact on said application. Because of the need for high power densities and/or effective heat transfer, applications such as cooling fabrics, air conditioning, electronics cooling, water heating, and battery thermal management stand to benefit significantly from AM-enabled advances. Given the size of TES needed, AM is unlikely to be a good fit for TES in solar thermal electricity applications.

Cost-effective mass manufacturability is key to the long-term commercial viability of AM PCM products. While initially an expensive technique, the cost of AM equipment has decreased significantly in the last few decades, with more reductions expected as the technology develops further.^[207] Prior studies on the industrial adoption of AM concluded that AM methods open design possibilities that are otherwise unattainable with traditional manufacturing techniques, allowing for significant cost reduction in the development of new optimized designs.^[208] Additional cost-relevant benefits include less material waste, lower labor costs, increased automation, and greater energy efficiency,^[208–210] painting a promising picture for the future of AM PCM devices.

One way to better understand the benefits and best applications for additively manufactured TES is through the use of Ragone plots.^[20,211–213] Ragone plots quantify the energy (kWh) versus power (kW) for PCM heat exchangers. A typical Ragone plot is shown in Figure 14a. The general goal of TES heat exchanger design is to maximize available energy (energy storage) and power (rate of heat transfer). In a Ragone plot, these goals

push the knee in the curve in Figure 14a further up and to the right. Achieving high energy and power densities requires low thermal resistances between the heat source or sink and the PCM phase front. Examples of how conductivity and geometry impact Ragone curves are shown in Figure 14a–c. These results were obtained using a 2D discretized model of a PCM heat exchanger, utilizing the enthalpy method, which neglects natural convection in the melt.^[20]

Increasing thermal conductivity lowers the resistance, which can increase the achievable power density from the baseline pure PCM curve (black) to the red curves that contain 2.5%, 5%, and 10% thermally conductive additives by volume. However, conductivity enhancement comes at the cost of reducing the volumetric storage capacity given that any scaffolding material displaces some of the active PCM. These effects must be considered together to achieve the optimal energy density for the desired charge or discharge rate (power), as shown in Figure 14c. The design goals are a function of the application. For applications that require relatively low charge/discharge rates, such as cold chain or building applications, effort should be focused on minimizing the weight and volume of the flow channels and scaffolding (decreasing length scales) rather than trying to increase the thermal conductivity through additives. Conversely, for applications that require high charge/discharge rates, such as electronics thermal management, focusing on thermal conductivity enhancement may be more beneficial.

An example of how geometry can impact the Ragone curve is shown in Figure 14b. These results specifically look at reducing the diameter of the fluid channel and increasing the total length, simulating a case where the fluid channels serpentine through the PCM. By modeling simple tubes (although the geometry is not optimized), Figure 14c shows how modifying the geometry

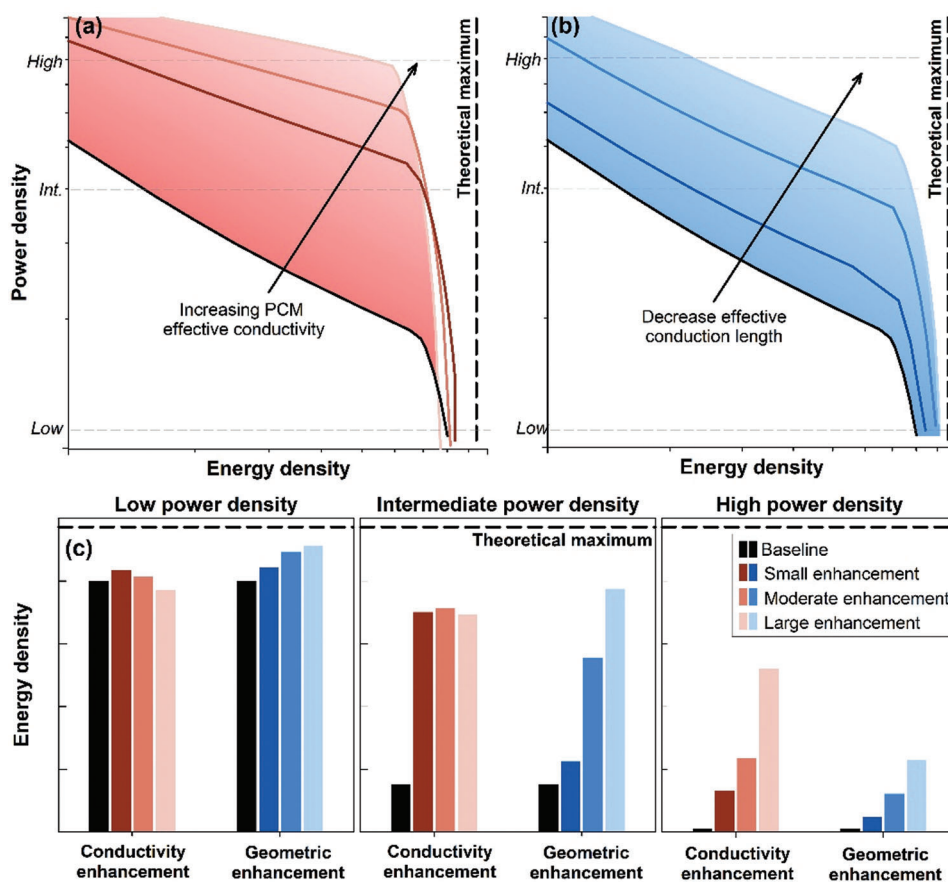


Figure 14. The performance of a thermal energy storage component in terms of energy and power density with different levels of enhancement, attainable using additive manufacturing. Panels (a) and (b) show Ragone plots for a round tube surrounded by PCM. The black curve shows the baseline case with a pure PCM, and the a) red and b) blue curves show different levels of conductivity and geometric enhancement, respectively. The energy density at three different power density levels (shown on the y-axis in (a) and (b)) is plotted for different enhancement schemes in (c). Note: The energy density is shown on the x-axis in the Ragone plots and on the y-axis in (c). The dashed black line in all three plots represents the maximum possible energy density, which is simply the energy storage capacity of the material alone with no flow channels or conductive additives.

by decreasing length scales can improve performance in all applications. The ability to design novel heat transfer-enhancing features with small length scales using AM for topology-optimized TES heat exchangers will make large gains in improving overall charge and discharge efficiencies. There exists a large body of research for AM that incorporates thermal conductivity enhancements; however, heat transfer within heat exchangers is largely driven by the surface area and the heat transfer coefficient that is achievable on the heat transfer fluid side, which is directly impacted by channel geometries. The optimization of additively manufactured heat exchangers, especially for TES, is still largely unexplored and should be a key area of future research. Further, as AM processes improve upon spatial resolution into the nanometer-to-micron regime, excessive PCM confinement may negatively impact TES properties and length-scale compromises may have to be made. As the field further develops, work toward design and demonstration of effective compact heat exchangers with high surface-area-to-volume ratios will showcase the potential for AM for TES applications.

Finally, thought should be given to the best current and future markets for AM TES devices. Today, one of the main advantages

of 3D printing (particularly extrusion-based) is that it enables fast, easy, cost-effective manufacturing without a full fabrication facility. This is especially helpful in situations where replacement parts are needed far from the original manufacturer (e.g., while in orbit in space or in a field location). It is expected that advanced manufacturing will play an important role in furthering the design of the heat exchanger form for TES systems. For additively manufactured TES to become competitive with traditional manufacturing techniques, the ability to scale up the technology from small form factors is imperative. Many 3D printers are prohibitively slow and can only produce very small footprints, making large heat exchangers that can be produced quickly nearly impossible. However, as additive manufacturing advances, the ability to produce compact, modular heat exchangers becomes more of a reality.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

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